SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

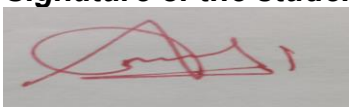
ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA0321	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	B.TECH-IT	**Date:**	09/10/2026

Code of Conduct:

I AATHISRI .P(192521252) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Instructions

- Topicgivenbeforethesessionandstudentsspeakon stage.
- Duration:10minutes(Inclass)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

INDIVIDUAL TASK: CELLULAR NETWORK

1. INTRODUCTION AND DEFINITION:

A **Cellular Network** is a wireless communication system in which a large geographical area is divided into smaller regions called **cells**. Each cell is served by a base station that provides wireless communication to mobile users.

Cellular networks allow users to make voice calls, send messages, and access the internet. They use the concepts of cell division, frequency reuse, and efficient resource allocation to provide communication over a large area.

2. CELL SHAPE:

A **cell** is a geographical area covered by a base station. In cellular network planning, a cell is commonly represented by a **hexagonal shape** because hexagons cover an area without gaps or overlaps in an idealized layout.

The common cell shapes are:

- **Hexagonal Cell:** Used in theoretical network planning because it provides a convenient representation of coverage.
- **Circular Cell:** Represents the approximate coverage area of an antenna, although actual radio coverage is irregular.
- **Square Cell:** Sometimes used in simplified models and simulations.

The hexagonal shape is preferred for theoretical analysis because it simplifies the calculation of coverage area and frequency reuse.

3. FREQUENCY REUSE:

Frequency Reuse is a technique in which the same set of radio frequencies is used in different cells that are sufficiently separated to limit interference.

In a cellular network, the available frequency spectrum is limited. Therefore, frequencies are reused in multiple cells to improve spectral efficiency and increase the number of users served.

The group of cells in which each frequency set is assigned once is called a **cluster**. After one cluster, the same frequency pattern can be repeated in another cluster.

The frequency reuse factor is given by:

Frequency Reuse Factor = 1 / N

Where:

- **N** = Number of cells in a cluster.

For example, if a cluster contains 7 cells, the frequency reuse factor is:

1/7

A smaller cluster can allow more frequent reuse of frequencies, but it may also increase co-channel interference.

4. CLUSTER:

A **Cluster** is a group of adjacent cells in which each available frequency set is assigned once before the pattern is repeated.

The cluster size is represented by **N** and is calculated using:

$$\mathbf{N = i^2 + ij + j^2}$$

Where:

- **i and j** are non-negative integers that determine the cell pattern.

Common cluster sizes include **3, 4, 7, 9, and 12**, depending on the selected reuse pattern and network design.

A suitable cluster size helps balance frequency reuse, interference, and network capacity.

5. CAPACITY OF A CELLULAR NETWORK:

Cellular Network Capacity refers to the number of users or communication channels that a cellular network can support at a given time while maintaining acceptable service quality.

The capacity depends on several factors:

- Available frequency spectrum
- Number of channels per cell
- Cluster size
- Frequency reuse factor
- Interference levels
- Cell size and coverage
- Number of base stations
- Available radio technologies

For a simplified FDMA/TDMA model, the number of channels per cell can be expressed as:

Channels per Cell = Total Available Channels / Cluster Size

For example, if 420 channels are available and the cluster size is 7:

Channels per Cell = 420 / 7 = 60 channels

Therefore, each cell receives 60 channels under this simplified allocation model.

6. HOW TO INCREASE CELLULAR NETWORK CAPACITY:

The capacity of a cellular network can be increased using the following techniques

1) Cell Splitting

A large cell is divided into smaller cells, each served by a base station. This allows frequencies to be reused more often and increases capacity in high-traffic areas.

2) Sectoring

A cell is divided into sectors using directional antennas. This helps reduce interference and can improve the reuse of available radio resources.

3) Microcells and Picocells

Small cells are installed in crowded locations such as shopping malls, railway stations, and urban areas to serve more users locally.

7. Conclusion:

To conclude, a **Cellular Network** is a wireless communication system that divides a geographical area into cells to provide efficient and reliable communication. The concepts of **cell shape, frequency reuse, cluster size, and network capacity** are important in cellular network design.

Frequency reuse allows the same frequencies to be used in different cells, while cluster size helps control interference and determine how frequently frequencies can be reused. Network capacity can be increased through cell splitting, sectoring, small cells, improved frequency planning, and advanced wireless technologies.

These techniques help cellular networks provide better coverage, higher capacity, improved spectral efficiency, and reliable communication for a growing number of mobile users.

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